

# ORDER FLOW SOFTWARE

## Functional Specification v3.0

*v3.0 adds: Section 7 — Bar/Candlestick Time Configuration | Section 8 — Timestamp Specification*

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May 2026 | Confidential & Proprietary

# 1. Complete Reference Registry

All sources consulted during the preparation of this specification — including the four PDF books provided by the user, the orderflows.com website, and all web sources independently researched — are catalogued below with their precise URLs.

## 1.1 Primary Source Documents (User-Supplied PDFs)

| # | Title                                                                                        | Author                 | Year | Coverage                                                                                                                                                                                                                                                                 |
|---|----------------------------------------------------------------------------------------------|------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Trading Order Flow: Understanding & Profiting From Market Generated Information As It Occurs | Michael Valtos         | 2015 | Footprint charts, Delta, COT, Imbalances, Stacked imbalance, Absorption, Trapped traders, Aggressive/passive participants, Initiative activity, VWAP, Iceberg orders, HFT, Candlestick + order flow, Chart types                                                         |
| 2 | ORDER FLOW: Trading Setups — Golden Ticket Edition                                           | Dale (trader-dale.com) | 2024 | Volume clusters, HVN, Multiple nodes, Trades filter, Stacked imbalances, Unfinished business, Confirmation setups (big limit orders, absorption, aggressive orders, CVD divergence), Take profit/stop loss with order flow, Volume profile shapes, Volume profile setups |
| 3 | VOLUME PROFILE: The Insider's Guide to Trading — Golden Ticket Edition                       | Dale (trader-dale.com) | 2024 | Price action strategies, Volume profile POC/VAH/VAL, D/P/b/thin profile shapes, Volume accumulation setup, Trend setup, Rejection setup, Money management, Position sizing, Backtesting phases, Trading psychology                                                       |
| 4 | VWAP Book (vwap.com)                                                                         | Dale (trader-dale.com) | 2024 | VWAP formula, Anchored VWAP (8 anchor types), VWAP strategies (reaction, rotation, trend), VWAP + Order Flow + Volume Profile confluences, Take profit placement, Stop loss placement, Trade trailing                                                                    |

## 1.2 Primary Website — orderflows.com (Newly Added)

 Fetched and fully read during preparation of v2.0. All pages listed below were directly accessed.

| Page / Section                                | URL                                                                                                                                                           | Key Content Extracted                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| orderflows.com Home                           | <a href="https://www.orderflows.com/">https://www.orderflows.com/</a>                                                                                         | Full product list, Pulse indicator summary, Turns indicator summary, Imbalance S&R summary, Single Prints summary, site navigation                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Orderflows Pulse                              | <a href="https://www.orderflows.com/pulse.html">https://www.orderflows.com/pulse.html</a>                                                                     | Full Pulse indicator specification: 7-variable composite (Order Flow, Delta, POC, Imbalances, Volume, Price Action, Swing Analysis) + Multi-Time Frame analysis upgrade, momentum jumps, order flow patterns                                                                                                                                                                                                                                                                                                                                            |
| Orderflows Turns                              | <a href="https://www.orderflows.com/turns.html">https://www.orderflows.com/turns.html</a>                                                                     | Full Turns indicator specification: POC, bid/ask volume, volume, price action, delta, swing analysis. Works on footprint AND regular bar charts. Identifies turning points.                                                                                                                                                                                                                                                                                                                                                                             |
| Imbalance Support & Resistance                | <a href="https://www.orderflows.com/2015/08/22/imbalance-support-and-resistance/">https://www.orderflows.com/2015/08/22/imbalance-support-and-resistance/</a> | Stacked imbalance zone indicator — draws buy/sell zones on chart, real-world analogy, 1–3 tick stops with 10x+ R/R potential                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Top Heavy Big Seller / Bottom Heavy Big Buyer | <a href="https://www.orderflows.com/2015/08/22/big-buyer-seller/">https://www.orderflows.com/2015/08/22/big-buyer-seller/</a>                                 | Orderflows Ratio indicator — detects institutional big buyer at low or big seller at high. Acts as price floor/ceiling. Identifies institutional hedges and spreads.                                                                                                                                                                                                                                                                                                                                                                                    |
| Single Prints – Last Buyer / Last Seller      | <a href="https://www.orderflows.com/2015/08/22/last-buyerseller/">https://www.orderflows.com/2015/08/22/last-buyerseller/</a>                                 | Single print box indicator — marks last buyer at high / last seller at low. Signals end of move. Works on short and medium term charts.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Delta Surge (Free)                            | <a href="https://www.orderflows.com/deltasurge/index.html">https://www.orderflows.com/deltasurge/index.html</a>                                               | Delta Surge indicator — analyzes delta to capitalize on real-time market information. Works on footprint AND normal bar charts. Free download. Delta trend analysis.                                                                                                                                                                                                                                                                                                                                                                                    |
| Orderflows Trader 7 Special                   | <a href="https://www.orderflows.com/special.html">https://www.orderflows.com/special.html</a>                                                                 | Full list of 30+ automatable signals: Accumulation/Distribution, Open POC, Aligned POCs, Orderflows Gaps, Imbalance Reload, Volume Decline, POC Wave, Delta Tail, Resting Liquidity, Vertical Liquidity, Delta Breakout, Retail Suck, Price Action Divergence, Prominent POC, Orderflows Tails, Delta Divergence, Exhaustion Prints, Imbalance Reversals, Market Sweep, Market Weakness, Orderflows Sequencing, POC Slingshot, Zero Print, Engulfing Value Area, Inverse Imbalance, Stacked Imbalance, Thin Prints, Orderflows Ratio, Price Exhaustion, |

|                                    |                                                                                                               |                                                                                                                                                                         |
|------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    |                                                                                                               | Price Defense, Multiple Imbalances                                                                                                                                      |
| Orderflows Inner Circle Video Club | <a href="https://www.orderflows.com/videoclub/index.html">https://www.orderflows.com/videoclub/index.html</a> | 50+ webinar topics: Bar Delta Divergence (Webinar 18), 2nd Slot Imbalances (Webinar 19), Valtos U-Turn indicator (Webinar 23), Chart Types (Webinar 49), and many more. |
| Orderflows Toolbox 8.0             | <a href="https://www.orderflows.com/toolbox/">https://www.orderflows.com/toolbox/</a>                         | Toolbox product — separate from NinjaTrader 8 basic Orderflows Toolkit. Lifetime license. No monthly fees.                                                              |
| Orderflows Software page           | <a href="https://www.orderflows.com/software/">https://www.orderflows.com/software/</a>                       | General software description, NinjaTrader integration details                                                                                                           |
| Markets Traded                     | <a href="https://www.orderflows.com/markets-traded/">https://www.orderflows.com/markets-traded/</a>           | Works on any futures market with adequate volume: ES, NQ, YM, CL, GC, ZB, 6E, grains, energy                                                                            |

### 1.3 Web Sources Independently Researched (Not Supplied by User)

***i** All URLs below were found and read during research sessions — not provided by the user. These are additional reference sources used to build the specification.*

| Source Name                                                          | URL                                                                                                                                                                                                                                   | What It Contributed                                                                                                                                             |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NinjaTrader Official Blog — Order Flow / Footprint Charts (Oct 2025) | <a href="https://ninjatrader.com/futures/blogs/ninjatrader-order-flow/">https://ninjatrader.com/futures/blogs/ninjatrader-order-flow/</a>                                                                                             | Official definition of Footprint Chart vs Volumetric Bars; Order Flow+ suite description; Lee-Ready classification mention; industry-standard term confirmation |
| TradeThePool — Footprint Charts Explained (Oct 2025)                 | <a href="https://tradethepool.com/fundamental/footprint-charts-explained/">https://tradethepool.com/fundamental/footprint-charts-explained/</a>                                                                                       | Visual definition of footprint profile; institutional interest zones; three-dimensional view of market activity                                                 |
| TradeThePool — Mastering Footprint Charts (Oct 2025)                 | <a href="https://tradethepool.com/fundamental/mastering-footprint-charts-trading/">https://tradethepool.com/fundamental/mastering-footprint-charts-trading/</a>                                                                       | Delta footprint definition; volume footprint definition; order flow footprint formal definition; unfinished auctions definition                                 |
| LiteFinance — Order Flow Trading with Footprint Charts (Apr 2026)    | <a href="https://www.litefinance.org/blog/for-beginners/trading-strategies/order-flow-trading-with-footprint-charts/">https://www.litefinance.org/blog/for-beginners/trading-strategies/order-flow-trading-with-footprint-charts/</a> | POC and Value Area definitions; passive vs aggressive order types; delta divergence as signal; delta trend analysis patterns                                    |
| OptFutures — Footprint Charts Complete Guide                         | <a href="https://optimusfutures.com/blog/footprint-charts/">https://optimusfutures.com/blog/footprint-charts/</a>                                                                                                                     | Three types of footprint charts (bid/ask, volume, delta); DOM explanation; stacked imbalances strategy; unfinished auctions strategy; image examples            |
| Wikipedia — Order Flow Trading                                       | <a href="https://en.wikipedia.org/wiki/Order_flow_trading">https://en.wikipedia.org/wiki/Order_flow_trading</a>                                                                                                                       | Buy/sell imbalance formal definition; spoofing/spoof orders definition; encyclopaedic reference for industry-                                                   |

|                                                                     |                                                                                                                                                                                                                       |                                                                                                                                                      |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                     |                                                                                                                                                                                                                       | standard terms                                                                                                                                       |
| Triad City Beat — Footprint Charts for Beginners (Jan 2026)         | <a href="https://triad-city-beat.com/footprint-charts-for-beginners-a-practical-guide-to-order-flow-trading/">https://triad-city-beat.com/footprint-charts-for-beginners-a-practical-guide-to-order-flow-trading/</a> | Platform list (Sierra Chart, NinjaTrader, Bookmap, MT5); retail trader access to institutional tools; row-level price level description              |
| TRADEPRO Academy — Footprint Chart Advantage                        | <a href="https://tradeproacademy.com/visualizing-market-order-flow-the-footprint-chart-advantage/">https://tradeproacademy.com/visualizing-market-order-flow-the-footprint-chart-advantage/</a>                       | Range bar footprint example; WHO is in control framework; ES E-mini footprint chart example                                                          |
| GoCharting — Orderflows Trader User Guide (Apr 2026)                | <a href="https://gocharting.com/docs/orderflow/orderflowtrader">https://gocharting.com/docs/orderflow/orderflowtrader</a>                                                                                             | Full Orderflows Trader platform documentation; stacked buying imbalance examples; delta candle divergence display; complete platform feature guide   |
| ClusterDelta — Footprint for cTrader                                | <a href="https://clusterdelta.com/ctrader-footprint">https://clusterdelta.com/ctrader-footprint</a>                                                                                                                   | Value Area Line Color settings; VAL/VAH line description; Profile BG inside/outside VA settings; cross-platform implementation reference             |
| ClusterDelta — Footprint for MetaTrader 4/5                         | <a href="https://clusterdelta.com/footprint">https://clusterdelta.com/footprint</a>                                                                                                                                   | Accumulation and distribution of market volumes; multi-platform compatibility reference; #Footprint indicator documentation                          |
| NordfX — Footprint Charts & Cumulative Delta (Nov 2025)             | <a href="https://nordfx.com/en/useful-articles/footprint-charts-cumulative-delta-order-flow-trading">https://nordfx.com/en/useful-articles/footprint-charts-cumulative-delta-order-flow-trading</a>                   | CVD divergence framework; combining footprint with classic price structure; demo account learning curve guidance; futures vs spot market correlation |
| Finowings — Order Flow Analysis: Footprint, Delta, Imbalance        | <a href="https://www.finowings.com/Trading/order-flow-analysis-footprint-delta">https://www.finowings.com/Trading/order-flow-analysis-footprint-delta</a>                                                             | Stacked imbalances support/resistance explanation; absorption definition; exhaustion definition; step-by-step order flow process for beginners       |
| Wikipedia — NinjaTrader                                             | <a href="https://en.wikipedia.org/wiki/NinjaTrader">https://en.wikipedia.org/wiki/NinjaTrader</a>                                                                                                                     | TASC Readers Choice Award history (2021–2024); Kraken acquisition May 2025 for \$1.5B; platform history and brokerage info                           |
| Silver Creek Leadership — Mike Valtos Ultimate Bundle 30 (Feb 2026) | <a href="https://silvercreekleadership.academy/courses/orderflows-mike-valtos-ultimate-bundle-30/">https://silvercreekleadership.academy/courses/orderflows-mike-valtos-ultimate-bundle-30/</a>                       | Course overview; 30 modules; 2025 volatility spike updates; community Telegram group; ongoing update schedule                                        |
| udcourse.com — Mike Valtos Bundle                                   | <a href="https://udcourse.com/product/mike-valtos-bundle-30-orderflows-courses/">https://udcourse.com/product/mike-valtos-bundle-30-orderflows-courses/</a>                                                           | 60–90 day screen time for tape reading mastery; Sierra Chart and NinjaTrader platform requirements; best markets (ES, NQ, CL, GC, ZB, BTC, ETH)      |
| Orderflows Inner Circle Group                                       | <a href="https://www.orderflows.com/innercircle/oldprice.html">https://www.orderflows.com/innercircle/oldprice.html</a>                                                                                               | Webinar 18 Bar Delta Divergence full definition; Webinar 19 2nd Slot Imbalances definition                                                           |
| orderflowstrader.com — Orderflows Trader 6                          | <a href="http://www.orderflowstrader.com/">http://www.orderflowstrader.com/</a>                                                                                                                                       | 8 new tools in v6: Thin Prints, Multiple Thin Prints, Delta/Volume Percentage, Extreme Delta/Volume, Extreme Delta,                                  |

|                                                          |                                                                                                                                                                               |                                                                                                                                                                         |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                          |                                                                                                                                                                               | Small Min/Max Delta                                                                                                                                                     |
| Orderflows Trader 7 Special page                         | <a href="https://www.orderflows.com/special.html">https://www.orderflows.com/special.html</a>                                                                                 | Complete automatable signal list for v7; Markers Plus compatibility; third-party automation framework                                                                   |
| Orderflows Toolbox page                                  | <a href="https://www.orderflows.com/toolbox/">https://www.orderflows.com/toolbox/</a>                                                                                         | Toolbox vs Toolkit distinction; lifetime license model; no monthly fees policy                                                                                          |
| govinfo.gov — Citadel CEO Congressional Testimony (VWAP) | <a href="https://www.govinfo.gov/content/pkg/CHRG-117hhrg43966/pdf/CHRG-117hhrg43966.pdf">https://www.govinfo.gov/content/pkg/CHRG-117hhrg43966/pdf/CHRG-117hhrg43966.pdf</a> | Official US Congressional testimony from Kenneth Griffin (Citadel CEO) confirming institutional VWAP program trading as virtually universal; VWAP slicing methodology   |
| flowshifts.com — Michael Valtos background               | <a href="http://www.flowshifts.com/">http://www.flowshifts.com/</a>                                                                                                           | Confirmed 20-year institutional career: 8yr JP Morgan, 4yr Cargill, 3yr Commerzbank, 2yr EDF Man, 4yr Dean Witter Reynolds; aggressive vs passive framework explanation |
| TradingIQ — OrderFlow IQ (TradingView)                   | <a href="https://fr.tradingview.com/script/GX4WhZ8h-TradingIQ-OrderFlow-IQ">https://fr.tradingview.com/script/GX4WhZ8h-TradingIQ-OrderFlow-IQ</a>                             | TradingView footprint implementation: POC, value area, imbalance detection, stacked imbalances, CVD levels across footprint                                             |

## 1.4 Reference Summary Count

| Category                           | Count | Details                                                                                                                                                                                                                                    |
|------------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| User-supplied PDF books            | 4     | Valtos Trading Order Flow; Dale Order Flow Setups; Dale Volume Profile; Dale VWAP                                                                                                                                                          |
| orderflows.com pages (newly added) | 12    | Home, Pulse, Turns, Imbalance S&R, Big Buyer/Seller, Single Prints, Delta Surge, Trader 7, Inner Circle, Toolbox, Software, Markets                                                                                                        |
| YouTube channel                    | 1     | @orderflows — channel metadata + video topic research                                                                                                                                                                                      |
| Independent web sources            | 23    | NinjaTrader, TradeThePool ×2, LiteFinance, OptFutures, Wikipedia ×2, Triad City Beat, TRADEPRO, GoCharting, ClusterDelta ×2, NordfX, Finowings, Silver Creek, udcourse, orderflowstrader, govinfo.gov, flowshifts, TradingIQ, Inner Circle |
| TOTAL                              | 40    | All sources catalogued above                                                                                                                                                                                                               |

## 2. orderflows.com Indicators — Full Specification

The following indicators are Michael Valtos's proprietary products sold at orderflows.com. They are directly relevant to the software we are building — they define the signals and features that the commercial product must detect and surface. All specifications below are derived from the orderflows.com website pages fetched directly.

### 2.1 Orderflows Pulse

Source: <https://www.orderflows.com/pulse.html> — fetched directly

Description: The Orderflows Pulse is Valtos's most comprehensive composite indicator. It analyzes 7 order flow variables simultaneously and generates trading signals on both footprint charts AND regular candlestick charts. It received a major upgrade with Multi-Time Frame analysis, allowing higher-timeframe order flow signals to be overlaid on lower-timeframe charts.

#### 2.1.1 The 7 Analysis Variables

| Variable               | What It Analyzes                                                            | Signal Contribution                                                      |
|------------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Order Flow             | What is currently trading and HOW it is trading in real-time                | Identifies order flow patterns and execution aggression                  |
| Delta                  | Net difference between aggressive buyers and aggressive sellers             | Shows which side is dominating the current bar                           |
| Point of Control (POC) | Price level with highest volume in the bar — only IMPORTANT POCs are used   | Identifies institutional consensus price; POC slingshot and defense      |
| Imbalances             | Price levels where aggressive traders outnumber others by large percentages | Detects supply/demand imbalance; generates buy/sell signals              |
| Volume                 | Sufficient volume threshold required for accurate assessment                | Filters noise bars; only signals on bars with meaningful volume          |
| Price Action           | How price is reacting to the order flow context                             | Confirms or negates the order flow signal with price behavior            |
| Swing Analysis         | Reference to recent significant swing highs and swing lows                  | Provides context for current order flow relative to key market structure |

#### 2.1.2 Multi-Time Frame Analysis (Major Upgrade)

- Allows higher-timeframe order flow signals to be displayed on the current chart without switching timeframes
- Example: trading a 30-second ES chart — add signals from the 3-minute ES chart on top
- Requires the same 7-variable analysis to pass on BOTH timeframes for highest-conviction signal
- Works on any chart type: minute-based, volume-based, tick-based, range-based

### 2.1.3 Pulse Signal Types

| Signal Type           | Definition                                                                                                | Trading Application                                                 |
|-----------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Momentum Jump         | Market momentum surges at a specific price — 'market sweet spot'. Optimal entry before and during trends. | Enter in direction of momentum jump; high-probability trend entry   |
| Order Flow Pattern    | Real-time analysis of how orders are executing — identifies immediate market dynamics                     | Confirms short-term directional bias based on live order activity   |
| Delta Imbalance       | Net aggressive buying vs selling dominance detected at the bar level                                      | Identifies which side controls the bar; filters setups by direction |
| Volume Concentration  | POC identification — highest volume price level in the bar                                                | Finds institutional activity center; used for entry/stop placement  |
| Price Imbalance       | Aggressive trader concentration significantly outnumbers opposing side at a price level                   | Identifies likely support/resistance from the order flow data       |
| Price Action Reaction | How price responds to the detected order flow conditions                                                  | Confirms or invalidates signal; filters false signals               |
| Swing Reference       | Context from recent swing highs and lows                                                                  | Gives market structure context to order flow signal                 |

### 2.1.4 Platform & Technical Details

- Runs on NinjaTrader 8 as an add-on indicator
- Install time: approximately 25 seconds
- Compatible with The Indicator Store's Markers Plus (third-party automation)
- Works on footprint charts (Orderflows Trader) AND regular bar/candlestick charts
- Works on minute, volume, tick, and range-based chart types with same settings
- Price: \$225 (one-time, 40% discount from \$397)

## 2.2 Orderflows Turns

Source: <https://www.orderflows.com/turns.html> — fetched directly

Description: The Orderflows Turns is designed specifically to identify market TURNING POINTS using a combination of 6 order flow and market analysis variables. It is simpler than the Pulse and more focused on reversal detection. Works on both footprint charts and normal bar/candlestick charts.

### 2.2.1 The 6 Analysis Variables

| Variable | What It Analyzes |
|----------|------------------|
|----------|------------------|



|                    |                                                                                          |
|--------------------|------------------------------------------------------------------------------------------|
| Point of Control   | Most-traded price within the bar — where institutional activity was concentrated         |
| Volume (Bid & Ask) | Volume traded on bid side vs ask side — detects aggressive order dominance               |
| Volume (Total)     | Sufficient total volume threshold to validate signal quality                             |
| Price Action       | How price is behaving relative to the order flow — confirms or denies the potential turn |
| Delta              | Net buyer vs seller aggression — is the turn confirmed by delta reversal?                |
| Swing Analysis     | Recent swing high/low reference points — is price at a significant structural level?     |

### 2.2.2 Turns Signal Behavior

- Identifies areas of the chart being defended or supported by institutional activity
- Like having Valtos 'looking over your shoulder' pointing out defended/supported areas
- NOT a simple price pattern — combines multiple order flow data points
- Can be overlaid on top of existing footprint charts OR existing candlestick charts
- Designed for short-term day traders: recommended 4–10 range charts, timeframes under 5 minutes to 30 seconds
- Works on: ES, CL, YM, NQ, 6E, ZB, FGBL and all major liquid futures markets
- Not recommended for thinly-traded markets (Lean Hogs, Live Cattle)
- Platform: NinjaTrader 8 version 15+. Price: \$500 one-time.

## 2.3 Orderflows Ratio — Top Heavy Big Seller / Bottom Heavy Big Buyer

Source: <https://www.orderflows.com/2015/08/22/big-buyer-seller/> — fetched directly

Description: The Orderflows Ratio indicator detects the presence of institutional players so large that their activity creates short-term price floors and ceilings. Based on Valtos's personal experience placing 1000+ lot institutional orders.

### 2.3.1 Signal Logic

- Bottom Heavy Big Buyer: detected at a LOW — a large institutional buyer absorbs all selling at a price level, acting as a price FLOOR
- Top Heavy Big Seller: detected at a HIGH — a large institutional seller satisfies all buyers at a price level, acting as a price CEILING
- Based on bid/ask RATIO across the bar — the ask/bid ratio exceeds a configured threshold indicating one side is massively dominant
- May represent commercial hedges or spread trades — Valtos's specific use case from his institutional trading career

### 2.3.2 Trading Application

- When Bottom Heavy Big Buyer appears at a low: buy the retest; stop below the detected level
- When Top Heavy Big Seller appears at a high: sell the retest; stop above the detected level
- 1–3 tick stops with profit potential 10x+ the risk (Valtos's stated specification)
- Works on the same markets as other Orderflows indicators — any liquid futures

## 2.4 Single Prints — Last Buyer / Last Seller

Source: <https://www.orderflows.com/2015/08/22/last-buyerseller/> — fetched directly

Description: The Single Print Box indicator identifies when the LAST buyer has bought at a high or the LAST seller has sold at a low. This marks the potential END of a trend leg — nobody wants to transact at that extreme anymore.

### 2.4.1 Signal Logic

- Last Buyer at High: only 1–2 trades executed at the high — demand has dried up at that price. Price will begin to fall.
- Last Seller at Low: only 1–2 trades at the low — supply has dried up at that price. Price will begin to rise.
- The single print (low/zero volume at extreme) is the visual indicator on the footprint chart
- Works as both scalp (short-term) and swing (medium-term) trade signal
- The indicator draws a box around the single print for easy visual identification

***i** This is the orderflows.com implementation of what Dale calls 'Unfinished Business' / 'Failed Auction'. Same concept, different naming convention.*

## 2.5 Delta Surge (Free Indicator)

Source: <https://www.orderflows.com/deltasurge/index.html> — fetched directly

Description: Delta Surge is a FREE indicator from Valtos that analyzes order flow delta and works on BOTH footprint charts AND normal bar charts. Its purpose is to help traders who are not yet using footprint charts to access delta analysis.

### 2.5.1 Signal Logic

- Analyzes delta trend — in a healthy uptrend, delta builds on itself as new buyers are drawn in
- Detects surges in delta that signal momentum acceleration in the prevailing direction
- Works independently of footprint chart — reads delta data from the instrument feed
- Allows delta analysis on standard candlestick charts without requiring Orderflows Trader
- Download: free at <https://www.orderflows.com/deltasurge/index.html>

- Use case: first step for traders transitioning from traditional charts to order flow analysis

## 2.6 Orderflows Indicators — Complete Feature Matrix for Implementation

The following table maps all orderflows.com indicators to required software features and signals the C++ engine must support:

| Indicator                           | Required Engine Signals                                                                      | Output to API                                              | Priority     |
|-------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------|--------------|
| Pulse                               | Delta, POC, Imbalances, Volume threshold, Price action context, Swing H/L, Multi-TF analysis | REST: /v1/signals/pulse<br>  WS: signals stream            | P1 —<br>v1.0 |
| Turns                               | POC, Bid/Ask ratio, Volume, Price action, Delta, Swing H/L, Turning point detection          | REST: /v1/signals/turns<br>  WS: signals stream            | P1 —<br>v1.0 |
| Orderflows Ratio (Big Buyer/Seller) | Bar-level bid/ask ratio exceeding threshold at high/low; volume confirmation                 | REST: /v1/signals/ratio  <br>WS: signals stream            | P1 —<br>v1.0 |
| Single Prints (Last Buyer/Seller)   | Zero or near-zero volume at bar extreme; single print box detection                          | REST:<br>/v1/signals/single_prints<br>  WS: signals stream | P1 —<br>v1.0 |
| Delta Surge                         | Delta trend acceleration; delta surge threshold detection; works on all chart types          | REST:<br>/v1/signals/delta_surge  <br>WS: delta stream     | P1 —<br>v1.0 |
| Stacked Imbalance Zones             | 3+ consecutive imbalances; zone drawing; persistence across bars                             | REST: /v1/zones   WS:<br>signals stream                    | P1 —<br>v1.0 |
| All Trader 7 signals (30+)          | Full signal set per Trader 7 spec (Accumulation, POC Wave, Delta Tail, etc.)                 | REST: /v1/signals   WS:<br>signals stream                  | P2 —<br>v1.5 |

## 3. Volume Profile as a Standalone Trading Instrument

Volume Profile is not merely a supporting indicator — it is a complete, standalone analytical instrument with its own data model, display specification, trading setups, and API surface. This section specifies Volume Profile as a first-class instrument in the software.

### 3.1 What Makes Volume Profile a Standalone Instrument

- Volume Profile provides the STRUCTURAL CONTEXT within which all order flow signals are interpreted
- Without Volume Profile, order flow signals have no reference frame — a stacked imbalance in the middle of a value area means something different from one at the POC
- Institutional traders rely on Volume Profile (Market Profile) to identify fair value, value area, and initiative vs responsive activity levels
- Dale explicitly states: 'I don't look for candlestick formations... I perceive the price movement as a continuous flow' — Volume Profile IS that continuous flow framework
- Volume Profile can be traded completely independently of order flow footprint charts — Dale's original approach before adding order flow

### 3.2 Volume Profile Data Model

| Data Element           | Type                   | Calculation                                           | Significance                                                                      |
|------------------------|------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------|
| POC — Point of Control | Price level            | Price with highest total volume in the profile period | Most-traded price = institutional consensus of fair value. Strongest S&R level.   |
| VAH — Value Area High  | Price level            | Upper boundary of 70% volume range above POC          | Upper edge of institutional comfort zone. Strong resistance.                      |
| VAL — Value Area Low   | Price level            | Lower boundary of 70% volume range below POC          | Lower edge of institutional comfort zone. Strong support.                         |
| Value Area             | Price range [VAL, VAH] | 70% of all traded volume falls within this range      | Zone of price acceptance. Two-sided trade. Most likely for price to return here.  |
| HVN — High Volume Node | Price cluster          | Volume significantly > average for adjacent levels    | Institutional accumulation/distribution zone. Strong S&R. Price spends time here. |
| LVN — Low Volume Node  | Price gap              | Volume significantly < average for adjacent levels    | Price acceptance gap. Price moves quickly through these levels.                   |
| Volume Distribution    | Histogram              | Volume at each price level plotted horizontally       | Full picture of institutional activity across all prices in                       |

|                       |                |                                                       |                                                                                         |
|-----------------------|----------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------|
|                       |                |                                                       | the period                                                                              |
| Bid/Ask Split Profile | Dual histogram | Ask volume (buy) vs Bid volume (sell) per price level | Shows directional bias within the profile — which side is more aggressive at each price |

### 3.3 Volume Profile Types (All Must Be Implemented)

| Profile Type                | Period Definition                           | Reset Behavior                 | Primary Use Case                                                  | Timeframe    |
|-----------------------------|---------------------------------------------|--------------------------------|-------------------------------------------------------------------|--------------|
| Session Profile             | Single RTH session (e.g., 9:30am–4:00pm ET) | Resets each session open       | Intraday trading — identifies today's fair value and value area   | 1min–30min   |
| Daily Profile               | One calendar day (including extended hours) | Resets at midnight             | Daily structure analysis; previous day POC as reference level     | 5min–1hr     |
| Weekly Profile              | Monday open to Friday close                 | Resets each Monday             | Swing trading context; weekly fair value benchmark                | 30min–4hr    |
| Monthly Profile             | Calendar month                              | Resets each month              | Institutional positioning; long-term supply/demand map            | Daily        |
| Yearly Profile              | January 1 to December 31                    | Resets each January 1          | Annual fair value; VWAP alignment analysis                        | Daily–Weekly |
| Flexible / Anchored Profile | User-defined start date to current bar      | Manual anchor — does not reset | Custom analysis: earnings events, regime changes, macro news      | Any          |
| Composite Profile           | Merged across multiple sessions             | Does not reset — accumulates   | Long-term institutional level mapping; multi-day absorption zones | Any          |
| Real-Time Profile           | Current session building in real time       | Continuous update each tick    | Live trading — watch POC migration, value area expansion          | Any          |

### 3.4 Volume Profile Shape Classification Engine

The engine must classify the current profile shape automatically and expose the classification via API. Shape determines the expected market behavior.

| Shape Name      | Visual Pattern | Mathematical Definition | Market Interpretation | Expected Next Move |
|-----------------|----------------|-------------------------|-----------------------|--------------------|
| D-Profile (Bell | Wide symmetric | Standard deviation of   | Balanced market. Two- | Continue           |

|                            |                                                       |                                                                               |                                                                                       |                                                                                |
|----------------------------|-------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Curve)                     | distribution centered on POC                          | volume distribution is low; POC is near the median price; tail volume < 15%   | sided trade. Neither buyers nor sellers dominating.                                   | rotation within value area until imbalance breaks out                          |
| P-Profile (Buy Imbalance)  | Narrow base, wide top; POC is near upper 30% of range | Volume skewed to upper prices; lower prices have thin volume tail             | Strong buying occurred near highs; accumulation at lows early, distribution later.    | Continuation UP if holding above POC; short on failure below POC               |
| b-Profile (Sell Imbalance) | Wide base, narrow top; POC is near lower 30% of range | Volume skewed to lower prices; upper prices have thin volume tail             | Strong selling occurred near lows; distribution at highs early, accumulation at lows. | Continuation DOWN if holding below POC; long on failure above POC              |
| Thin Profile               | Narrow, elongated; minimal volume at most levels      | Standard deviation of volume distribution is high; no clear POC concentration | Trending/initiative market. Price moved quickly. Little consensus.                    | Fast reversion to value if move is responsive; fast continuation if initiative |

### 3.5 Volume Profile Trading Setups (Standalone)

Source: Dale — Volume Profile Insider's Guide; Dale — Order Flow Trading Setups ch. Volume Profile

#### Setup VP-1: Volume Accumulation Setup (D-Profile Breakout)

- Identify a D-profile (balanced, bell-curve) area — price has been rotating inside value area
- Wait for price to break OUT of the value area with conviction (bar closing clearly above VAH or below VAL)
- Entry trigger: first bar closing outside value area with delta 3x or more above average
- Stop: back inside the value area (VAH becomes support for long; VAL becomes resistance for short)
- Target: next significant HVN, POC, or VWAP level beyond the breakout
- Logic: the D-profile represents institutional accumulation — breakout signals their distribution/initiative

#### Setup VP-2: Trend Setup (P/b-Profile Continuation)

- Identify a P-profile (bullish) or b-profile (bearish) — shows directional institutional intent
- P-profile: wait for pullback to POC or VAH from above; look for long entry
- b-profile: wait for rally to POC or VAL from below; look for short entry

- Entry trigger: delta confirmation at the pullback level; delta should be in trend direction
- Stop: clearly on wrong side of POC (below for longs in P-profile; above for shorts in b-profile)
- Target: previous swing high/low; next week's VWAP; next profile POC
- Logic: POC is institutional consensus — they will defend it in trend direction

### **Setup VP-3: Rejection Setup (Thin Profile Reversal)**

- Identify a thin profile area — price moved quickly through with little volume = no institutional acceptance
- Price reaches the thin profile extreme and shows aggressive rejection delta
- Entry trigger: rejection bar with strong opposing delta; CVD divergence at thin profile extreme
- Stop: beyond the thin profile extreme (1–2 ticks past the zero-volume zone)
- Target: back to the origin of the thin profile (nearest HVN or previous session POC)
- Logic: institutions did not transact here — no support/resistance. Price snaps back to value.

### **Setup VP-4: Value Area Re-Entry (Responsive Trade)**

- Price breaks BELOW VAL or ABOVE VAH but with weak or negative delta — responsive move
- Price rapidly reverses back inside the value area
- Entry: on return into value area at VAL/VAH with confirming order flow
- Stop: beyond the recent extreme (outside the value area)
- Target: POC (value area center) — responsive trades target a return to fair value
- Logic: institutions did not support the breakout — they faded it with limit orders

### **Setup VP-5: POC as Dynamic Support/Resistance**

- The POC of the previous session/day/week acts as a reference level for the next session
- Previous Session POC (VPOC): if current price is above, acts as support; if below, acts as resistance
- Strong signal when current session volume builds around the previous VPOC — institutional memory
- Entry: test of previous VPOC with order flow confirmation (absorption or buying imbalance at support VPOC)
- Stop: break below previous VPOC with strong sell delta
- Target: current session VAH or VAL

### **Setup VP-6: Volume Profile + Order Flow Confluence (Highest Conviction)**

- Find a Volume Profile key level (POC, VAH, VAL, HVN) — this is the structural setup

- Wait for price to arrive at the level
- Require order flow confirmation: stacked imbalance OR absorption OR delta divergence at the level
- Also require VWAP alignment: price at VP level that also coincides with Daily or Weekly VWAP
- Triple confluence (VP + Order Flow + VWAP) = highest conviction trade signal in the system
- Entry: at the confluence level; stop 1–3 ticks outside; target 2:1 or better R/R to next structure

***i** This is the core methodology that both Valtos and Dale describe as their preferred approach. Three-way confluence is the gold standard.*

### 3.6 Volume Profile API Surface

| Endpoint                              | Method     | Parameters                                        | Response                                                                 |
|---------------------------------------|------------|---------------------------------------------------|--------------------------------------------------------------------------|
| /v1/profile/{instrument_id}           | GET        | period_type, start_date, end_date, session_filter | { poc, vah, val, value_area_pct, hvns[], lvns[], shape, distribution[] } |
| /v1/profile/{instrument_id}/live      | GET        | period_type                                       | Real-time current session profile building                               |
| /v1/profile/{instrument_id}/composite | GET        | from_date, to_date                                | Composite profile across multiple sessions                               |
| /v1/profile/{instrument_id}/anchored  | GET        | anchor_timestamp                                  | Flexible anchored profile from any point to current bar                  |
| WS: profile/{instrument}              | WebSoc ket | Subscribe to profile type                         | Live updates: { poc, vah, val, shape, updated_at } on each bar close     |

### 3.7 Volume Profile Display Specification

| Display Element       | Default Color                                        | Configurable | Notes                                                    |
|-----------------------|------------------------------------------------------|--------------|----------------------------------------------------------|
| Volume histogram bars | Blue gradient (ask-heavy) / Red gradient (bid-heavy) | Yes          | Darker = higher volume. Optional bid/ask split coloring. |
| POC line              | Yellow horizontal line across chart                  | Yes          | Most prominent line on the profile                       |
| VAH line              | Green dashed horizontal line                         | Yes          | Upper value area boundary                                |
| VAL line              | Red dashed horizontal line                           | Yes          | Lower value area boundary                                |
| Value area fill       | Semi-transparent blue                                | Yes          | Shaded region between VAL and VAH                        |



|                    |                                       |     |                                                |
|--------------------|---------------------------------------|-----|------------------------------------------------|
| HVN markers        | Bold/wider volume bar                 | Yes | HVN automatically detected and highlighted     |
| LVN markers        | Thin/barely visible gap               | Yes | LVN gaps automatically highlighted             |
| Shape label        | Text overlay: D / P / b / Thin        | Yes | Auto-classified shape label displayed on chart |
| Profile type label | Text: Session / Daily / Weekly / etc. | Yes | Shows which profile type is displayed          |

### 3.8 Volume Profile Configuration Parameters

| Parameter                | Default | Range                                                                          | Description                                                         |
|--------------------------|---------|--------------------------------------------------------------------------------|---------------------------------------------------------------------|
| value_area_percent       | 70      | 60–90%                                                                         | Percentage of volume defining value area (academic standard is 70%) |
| profile_type             | SESSION | SESSION/<br>DAILY/<br>WEEKLY/<br>MONTHLY/<br>YEARLY/<br>FLEXIBLE/<br>COMPOSITE | Which profile type to display                                       |
| show_multiple_profiles   | false   | true/false                                                                     | Show multiple profiles simultaneously (e.g., session + weekly)      |
| hvn_threshold_multiplier | 1.5     | 1.1–3.0                                                                        | Volume must exceed average × multiplier to qualify as HVN           |
| lvn_threshold_multiplier | 0.5     | 0.1–0.9                                                                        | Volume must be less than average × multiplier to qualify as LVN     |
| bid_ask_split            | false   | true/false                                                                     | Show bid and ask volume separately in profile histogram             |
| profile_width_pct        | 20      | 5–50%                                                                          | Profile histogram width as percentage of chart width                |
| auto_classify_shape      | true    | true/false                                                                     | Auto-detect and label D/P/b/Thin profile shape                      |
| show_previous_poc        | true    | true/false                                                                     | Display previous session/period POC as reference line               |
| composite_lookback_days  | 20      | 5–252                                                                          | Number of days to include in composite profile                      |
| real_time_update_ms      | 100     | 100–5000ms                                                                     | How often live profile recalculates during session                  |

## 4. Updated Complete Feature Matrix

Full list of all features across Order Flow, Volume Profile, VWAP, and orderflows.com Indicators that the software engine must implement.

| Feature Category | Feature                                                          | Source                              | Version |
|------------------|------------------------------------------------------------------|-------------------------------------|---------|
| Footprint        | Bid/Ask volume per price level                                   | Valtos book; Dale book; NinjaTrader | v1.0    |
| Footprint        | Cell color (green/red) by ask vs bid dominance                   | Valtos; Dale                        | v1.0    |
| Footprint        | Volume heat map (darker = higher)                                | Dale                                | v1.0    |
| Footprint        | Diagonal imbalance comparison                                    | Valtos; Dale                        | v1.0    |
| Delta            | Bar delta (ask_vol - bid_vol)                                    | Valtos; Dale; GoCharting            | v1.0    |
| Delta            | Max delta / Min delta within bar                                 | Valtos                              | v1.0    |
| Delta            | Cumulative Delta (CVD) — session total                           | Dale; GoCharting                    | v1.0    |
| Delta            | Delta/Volume percentage                                          | Orderflows Trader 6                 | v1.0    |
| Delta            | Extreme delta detection                                          | Orderflows Trader 6                 | v1.0    |
| Delta            | Small Min/Max delta detection                                    | Orderflows Trader 6                 | v1.0    |
| Delta            | Delta Surge indicator                                            | orderflows.com/deltasurge           | v1.0    |
| Delta Divergence | Standard: negative delta at new high / positive at new low       | Valtos; Dale                        | v1.0    |
| Delta Divergence | Bar Delta Divergence: bearish bar + positive delta               | Valtos Webinar 18                   | v1.0    |
| Delta Divergence | CVD Divergence: price vs cumulative delta                        | Dale                                | v1.0    |
| Imbalance        | Single buying/selling imbalance (2.5x/3x/4x)                     | Valtos; Dale; OptFutures            | v1.0    |
| Imbalance        | Stacked imbalance (3+) zone with zone drawing                    | Valtos; Dale; orderflows.com        | v1.0    |
| Imbalance        | 2nd Slot imbalance detection                                     | Valtos Webinar 19                   | v1.0    |
| Imbalance        | Imbalance reversal                                               | Orderflows Trader 7                 | v1.0    |
| Imbalance        | Multiple imbalances cluster                                      | Orderflows Trader 6/7               | v1.0    |
| Imbalance        | Inverse imbalance                                                | Orderflows Trader 6                 | v1.0    |
| Imbalance        | Imbalance reload                                                 | Orderflows Trader 7                 | v1.0    |
| COT              | Intrabar COT: highest volume price level in bar with box display | Valtos book                         | v1.0    |

|                           |                                                                     |                                 |      |
|---------------------------|---------------------------------------------------------------------|---------------------------------|------|
| Absorption                | High volume + low net delta at price level                          | Valtos; Dale                    | v1.0 |
| Trapped Traders           | Imbalance at bar extreme in opposite direction to bar close         | Valtos book                     | v1.0 |
| Initiative/<br>Responsive | Price vs value area + delta at breakout                             | Valtos book                     | v1.0 |
| Single Prints             | Last buyer at high / last seller at low detection                   | orderflows.com/last-buyerseller | v1.0 |
| Orderflows Ratio          | Top heavy big seller / bottom heavy big buyer                       | orderflows.com/big-buyer-seller | v1.0 |
| Orderflows Pulse          | 7-variable composite signal + multi-TF analysis                     | orderflows.com/pulse.html       | v1.0 |
| Orderflows Turns          | 6-variable turning point detector                                   | orderflows.com/turns.html       | v1.0 |
| Volume Profile            | Session profile with POC/VAH/VAL/HVN/LVN                            | Dale VP book; orderflows.com    | v1.0 |
| Volume Profile            | Daily/Weekly/Monthly/Yearly profiles                                | Dale VP book                    | v1.0 |
| Volume Profile            | Flexible / Anchored profile                                         | Dale VP book                    | v1.0 |
| Volume Profile            | Composite profile (multi-session)                                   | Dale VP book                    | v1.0 |
| Volume Profile            | D/P/b/Thin shape auto-classification                                | Dale VP book                    | v1.0 |
| Volume Profile            | Prominent POC + Aligned POCs + Open POC                             | Orderflows Trader 7             | v1.0 |
| Volume Profile            | POC Wave + POC Slingshot                                            | Orderflows Trader 7             | v1.0 |
| Volume Profile            | Bid/Ask split profile histogram                                     | Dale book                       | v1.0 |
| Volume Profile            | VP Setup 1–6 signal detection                                       | Dale; this spec §3.5            | v1.0 |
| VWAP                      | Daily/Weekly/Yearly VWAP                                            | Dale VWAP book                  | v1.0 |
| VWAP                      | Anchored VWAP (8 anchor types)                                      | Dale VWAP book                  | v1.0 |
| VWAP                      | Standard deviation bands ( $1\sigma$ , $2\sigma$ )                  | Dale VWAP book                  | v1.0 |
| VWAP                      | VWAP reaction/rotation/trend setups                                 | Dale VWAP book                  | v1.0 |
| Trader 7 Signals          | All 30+ automatable signals (Accumulation thru Multiple Imbalances) | Orderflows Trader 7 spec        | v1.5 |
| Market Structure          | Session high/low, previous day high/low                             | Valtos book                     | v1.0 |
| Market Structure          | Unfinished Business / Failed Auction detector                       | Dale book; orderflows.com       | v1.0 |
| Chart Types               | TIME / VOLUME / RANGE bar construction                              | Valtos book                     | v1.0 |

## 5. Implementation Notes — v2.0 Additions

### 5.1 Volume Profile as Primary Data Layer

Volume Profile data must be calculated and available BEFORE order flow signals are evaluated. The signal engine depends on VP context for signal validity:

- Stacked imbalance at VAH is stronger than stacked imbalance in middle of value area
- Delta divergence AT the POC is a high-conviction signal; delta divergence away from POC is lower conviction
- Absorption at the POC has different meaning than absorption at an HVN vs an LVN
- All 6 VP trading setups must be cross-referenced with order flow signals before alerting the user

### 5.2 orderflows.com Indicator Compatibility

 *The Pulse and Turns indicators are sold as commercial NinjaTrader products by Valtos. We are building our OWN implementation of these signal types, not reverse-engineering his protected code. Our signals follow the same analytical logic as documented on orderflows.com.*

- Pulse implementation: combine all 7 variables in a weighted scoring engine; threshold score triggers signal
- Turns implementation: 6-variable analysis with turning point pattern recognition
- Ratio implementation: bar-level bid/ask ratio at price extremes with configurable threshold
- Single Prints implementation: zero-print / last-print detection at bar highs/lows
- Delta Surge implementation: delta acceleration detection using rolling delta window

### 5.3 Multi-Instrument Volume Profile

Per Valtos's teaching: the same Volume Profile concepts apply across ALL futures markets. The software must support configuring profiles for:

- Index futures: ES (S&P 500), NQ (Nasdaq 100), YM (Dow Jones), RTY (Russell 2000)
- Energy: CL (Crude Oil), NG (Natural Gas), RB (RBOB Gasoline)
- Metals: GC (Gold), SI (Silver), HG (Copper)
- Fixed income: ZB (30-yr T-Bond), ZN (10-yr T-Note), ZF (5-yr T-Note)
- FX futures: 6E (Euro), 6B (British Pound), 6J (Japanese Yen), 6A (Australian Dollar)
- Agricultural: ZC (Corn), ZS (Soybeans), ZW (Wheat)
- Crypto futures: BTC, ETH (per udcourse.com reference)

 *Volume Profile parameters (value\_area\_percent, HVN/LVN thresholds) may need instrument-specific tuning — ES has different volume distribution characteristics than CL or ZB.*

## 5.4 Revised Implementation Priority

| Phase                            | Timeline   | Key Additions from v2.0                                                                           |
|----------------------------------|------------|---------------------------------------------------------------------------------------------------|
| Phase 0: Interfaces              | Week 1     | Add Volume Profile module headers; add orderflows.com indicator interface headers                 |
| Phase 1: Core Engine             | Week 2–4   | Tick ingestion, bar construction, delta engine, basic footprint — unchanged                       |
| Phase 2: Volume Profile Engine   | Week 4–6   | MOVED UP: Volume Profile is now Phase 2 (was Phase 3) — required as foundation for signal context |
| Phase 3: Order Flow Analytics    | Week 6–8   | Full imbalance suite, absorption, COT, trapped traders — now uses VP context                      |
| Phase 4: orderflows.com Signals  | Week 8–10  | Pulse, Turns, Ratio, Single Prints, Delta Surge implementations                                   |
| Phase 5: VWAP + Advanced Signals | Week 10–12 | VWAP engine; all VWAP strategies; Trader 7 signals subset                                         |
| Phase 6: API Layer               | Week 12–14 | REST + WebSocket + VP API surface + orderflows.com signal endpoints                               |
| Phase 7: NinjaTrader + Testing   | Week 14–18 | NinjaTrader adapter; full test suite; historical replay                                           |
| Phase 8: Security + Launch       | Week 18–22 | License engine; obfuscation; Paddle integration; NinjaTrader v1.0 public release                  |

## 6. Additional Glossary Entries (v2.0)

| Term                                    | Definition                                                                                                                                                                                                                             | Source                                                                               |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Orderflows Pulse                        | Composite order flow indicator analyzing 7 variables (Order Flow, Delta, POC, Imbalances, Volume, Price Action, Swing Analysis) with multi-timeframe capability. Generates momentum jump, delta imbalance, and price reaction signals. | <a href="http://orderflows.com/pulse.html">orderflows.com/pulse.html</a>             |
| Orderflows Turns                        | Order flow turning point detector analyzing 6 variables (POC, Bid/Ask volume, Volume, Price Action, Delta, Swing Analysis). Identifies defended/supported price areas for reversal entries.                                            | <a href="http://orderflows.com/turns.html">orderflows.com/turns.html</a>             |
| Orderflows Ratio                        | Indicator detecting institutional-scale buyers (Bottom Heavy) at lows or sellers (Top Heavy) at highs — based on bid/ask ratio at price extremes.                                                                                      | <a href="http://orderflows.com/big-buyer-seller">orderflows.com/big-buyer-seller</a> |
| Single Print / Last Buyer / Last Seller | Zero or near-zero volume at a bar extreme — signals the last trade at that price. Marks potential end of move. Orderflows implementation of Unfinished Business concept.                                                               | <a href="http://orderflows.com/last-buyerseller">orderflows.com/last-buyerseller</a> |
| Delta Surge                             | Free indicator detecting surges in delta trend. Works on both footprint and regular bar charts. Signals momentum acceleration.                                                                                                         | <a href="http://orderflows.com/deltasurge">orderflows.com/deltasurge</a>             |
| Bottom Heavy                            | Bar pattern where the majority of volume and bid/ask activity concentrates at the LOWER price levels — indicates strong passive or aggressive buying at lows.                                                                          | <a href="http://orderflows.com/big-buyer-seller">orderflows.com/big-buyer-seller</a> |
| Top Heavy                               | Bar pattern where the majority of volume and bid/ask activity concentrates at the UPPER price levels — indicates strong passive or aggressive selling at highs.                                                                        | <a href="http://orderflows.com/big-buyer-seller">orderflows.com/big-buyer-seller</a> |
| Momentum Jump                           | Pulse signal: sudden acceleration in order flow momentum at a specific price — identified as optimal entry point.                                                                                                                      | <a href="http://orderflows.com/pulse.html">orderflows.com/pulse.html</a>             |
| Standalone Volume Profile               | Volume Profile used as the primary trading instrument rather than a supporting indicator. Full analytical framework including profile types, shape classification, and 6 trading setups.                                               | This spec §3; Dale VP book                                                           |
| D-Profile                               | Bell-curve shaped volume profile — balanced, rotational market with two-sided trade facilitation.                                                                                                                                      | Dale Volume Profile book                                                             |
| P-Profile                               | Volume profile with concentration at upper prices (narrow base, wide top) — bullish accumulation/distribution pattern.                                                                                                                 | Dale Volume Profile book                                                             |
| b-Profile                               | Volume profile with concentration at lower prices (wide base, narrow top) — bearish distribution/accumulation pattern.                                                                                                                 | Dale Volume Profile book                                                             |

|                      |                                                                                                                                            |                                |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| Triple Confluence    | Highest-conviction setup: Volume Profile key level + Order Flow signal (imbalance/absorption) + VWAP level all aligning at the same price. | Dale VWAP book; this spec §3.5 |
| Value Area Re-Entry  | Trade setup: price breaks out of the value area on weak/responsive delta and rapidly returns inside — fade the breakout.                   | This spec §3.5<br>Setup VP-4   |
| Previous Session POC | The POC from the prior session that carries forward as a reference level for the current session — institutional memory level.             | This spec §3.5<br>Setup VP-5   |

## 7. Bar & Candlestick Time Configuration

Every bar (candlestick) in the Order Flow Engine is fully time-configurable. 'Bar' and 'candlestick' refer to the same unit of market data — the difference is only visual. A bar shows OHLC as lines; a candlestick shows the same data as a coloured body with wicks. The underlying data is identical and all order flow metrics (delta, imbalance, footprint, COT) are computed identically regardless of which visual style is selected.

### 7.1 Bar Construction Methods

The engine supports three fundamentally different methods of constructing bars. The choice determines WHEN a new bar opens and closes:

| Method | New Bar Opens When...                                                 | New Bar Closes When...                                      | Valtos Preference    | Best For                                                                                    |
|--------|-----------------------------------------------------------------------|-------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------------|
| TIME   | A clock boundary is crossed (e.g. every 5 minutes on the minute)      | The next clock boundary arrives                             | 3rd choice           | Alignment with economic events; most trader-familiar; easy to compare sessions              |
| RANGE  | The previous bar's price range is complete (e.g. 4 ticks of movement) | Price moves outside the configured tick range from bar open | 1st choice (primary) | Volatility-adaptive; new bars only form when price MOVES; clean order flow signal isolation |
| VOLUME | The previous bar accumulated the configured contract count            | The running volume total reaches the configured threshold   | 2nd choice           | Equal-volume bars; institutional activity analysis; size-normalised footprints              |

***i** Valtos explicitly states in the source book: 'Given a choice between range based charts, volume based charts and time based charts I would choose range based charts followed by volume charts followed by time based charts.' However all three methods are fully configurable by the user.*



## 7.2 TIME-Based Bar Sizes — Complete Configuration Table

For TIME-based bars, the user selects a time duration. Every valid duration the engine supports is listed below across three tiers: sub-minute (seconds), minute-based, and hour/session-based.

### 7.2.1 Sub-Minute (Seconds) — Tier 1

| Bar Size   | Config Value | Bars Per RTH Session (6.5hr) | Typical Use Case                                        | Order Flow Notes                                                                                                                        |
|------------|--------------|------------------------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 5 seconds  | 5s           | 4,680                        | High-frequency scalping; HFT pattern analysis           | Extremely noisy footprint. Only useful for liquid markets (ES, NQ) with 50,000+ contracts/session. Delta very small per bar.            |
| 10 seconds | 10s          | 2,340                        | Scalping; short-term momentum; tape reading             | Still noisy but manageable. Begin to see meaningful imbalances at high-volume moments.                                                  |
| 15 seconds | 15s          | 1,560                        | Scalping; fast intraday momentum                        | Better imbalance signal quality. Pulse and Turns signals become more reliable than 5s/10s.                                              |
| 20 seconds | 20s          | 1,170                        | Scalping; short-term delta divergence                   | Less common but valid. Use when 15s is too noisy and 30s is too slow.                                                                   |
| 30 seconds | 30s          | 780                          | Active scalping; Valtos recommended for Turns indicator | Good balance for scalpers. Enough ticks per bar for meaningful COT and imbalance detection. Valtos specifically mentions 30s for Turns. |
| 45 seconds | 45s          | 520                          | Scalping to short-term day trading                      | Uncommon. Most traders skip to 1min. Valid if 30s is too fast and 1min is too slow.                                                     |

### 7.2.2 Minute-Based — Tier 2 (Most Common for Day Trading)

| Bar Size  | Config Value | Bars Per RTH Session | Typical Use Case                                                            | Order Flow Notes                                                                                                         |
|-----------|--------------|----------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 1 minute  | 1m           | 390                  | Short-term day trading; delta divergence watch; Pulse signals on fast moves | Dale's preferred TF for watching CVD divergence. Enough volume for meaningful delta. Stacked imbalances become reliable. |
| 2 minutes | 2m           | 195                  | Day trading; bridge between 1min and 3min                                   | Less common. Use when 1min is too choppy. Imbalances and COT very clear.                                                 |

|                     |           |     |                                                                                                                  |                                                                                                                                                                                                                         |
|---------------------|-----------|-----|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 minutes           | 3m        | 130 | Day trading; Pulse multi-TF confirmation (use as higher TF for 30s primary)                                      | Valtos mentions 3-minute as the higher TF for Pulse multi-TF analysis when trading 30-second charts.                                                                                                                    |
| 4 minutes           | 4m        | 97  | Uncommon; transitional size                                                                                      | Rarely used in practice. Valid configuration.                                                                                                                                                                           |
| 5 minutes           | 5m        | 78  | Primary intraday chart (most popular). Dale's preferred for VWAP analysis. Orderflows Pulse default.             | Industry standard for order flow analysis. Dale explicitly states '5-minute is my personal preference for Daily VWAP trading.' Valtos uses 5min for many of his book examples. Best balance of signal quality vs noise. |
| 6 minutes           | 6m        | 65  | Uncommon; used by some prop traders                                                                              | Valid but rare. Divides the RTH session into exact 65 bars.                                                                                                                                                             |
| 10 minutes          | 10m       | 39  | Medium-term day trading; swing trade entry timing                                                                | Fewer, cleaner bars. Stacked imbalance zones more meaningful. Good for higher-timeframe context.                                                                                                                        |
| 12 minutes          | 12m       | 32  | Uncommon                                                                                                         | Valid. Used rarely.                                                                                                                                                                                                     |
| 15 minutes          | 15m       | 26  | Stacked imbalance hunting (Valtos recommendation); swing trade context; Orderflows Inner Circle webinar examples | Valtos explicitly recommends 15-minute charts for finding stacked imbalances: 'I like to look for these trades on longer time frame charts like a 15 minute or 30 minute chart.' Strong institutional signals.          |
| 20 minutes          | 20m       | 19  | Intermediate term; pre-market analysis                                                                           | Clear, high-conviction bars. HVN and POC levels very significant at this TF.                                                                                                                                            |
| 30 minutes          | 30m       | 13  | Swing trade context; Dale's preferred for Weekly VWAP; session structure analysis                                | Dale explicitly: 'for intraday trading with a 30-minute time frame... I find it more effective than the Daily VWAP.' Very clean Volume Profile and VWAP signals.                                                        |
| 45 minutes          | 45m       | 8   | Near-session-level analysis                                                                                      | Uncommon but valid. 8 bars per session gives macro view.                                                                                                                                                                |
| 60 minutes / 1 hour | 60m or 1h | 6   | Session structure; institutional bias; swing trade entry                                                         | 6 bars per RTH session. Each bar = 1 hour of institutional activity. Very clear initiative vs responsive                                                                                                                |

|  |  |  |  |          |
|--|--|--|--|----------|
|  |  |  |  | context. |
|--|--|--|--|----------|

### 7.2.3 Hour and Session Based — Tier 3

| Bar Size | Config Value | Bars Per Session / Day     | Typical Use Case                                      | Order Flow Notes                                                                                                |
|----------|--------------|----------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 2 hours  | 2h           | 3 per RTH session          | Macro intraday bias; open/midday/close structure      | Each bar represents a major session segment. Delta and Volume Profile per segment shows institutional rotation. |
| 3 hours  | 3h           | 2 per RTH session          | Open vs afternoon analysis                            | 2 bars: open drive + afternoon. Clear initiative vs responsive separation.                                      |
| 4 hours  | 4h           | ~1.6 per RTH session       | Swing trading; daily bias chart                       | Standard for swing traders alongside daily chart.                                                               |
| 6 hours  | 6h           | 1 per RTH session (approx) | Full-session single-bar; macro institutional view     | Entire RTH session in one bar. Delta tells you if the day was net buying or selling.                            |
| Daily    | 1D           | 1 per calendar day         | Swing trading; position trading; yearly VWAP analysis | One bar per day. Order flow shows daily institutional bias. Volume Profile becomes the session profile.         |
| Weekly   | 1W           | 1 per 5 trading days       | Position trading; long-term S/R identification        | Weekly bars with order flow. Rare but powerful for identifying multi-week imbalance zones.                      |
| Monthly  | 1MO          | ~21 trading days           | Long-term investment; yearly institutional mapping    | Monthly footprints. Mostly for research and institutional-grade analysis.                                       |

## 7.3 RANGE-Based Bar Sizes

For RANGE bars, the user specifies the number of TICKS (minimum price increments) each bar spans. A new bar opens only when price moves outside the previous bar's range. Bars may take seconds or minutes to complete depending on volatility.

| Instrument          | Tick Size         | Common Range Sizes               | Notes                                                                                       |
|---------------------|-------------------|----------------------------------|---------------------------------------------------------------------------------------------|
| ES (E-mini S&P 500) | \$12.50 / 0.25pts | 2, 4, 6, 8, 10, 12, 16, 20 ticks | Valtos uses 4-range as his default in many examples. 4 range = 1 point = \$50 per contract. |
| NQ (E-mini Nasdaq)  | \$5.00 / 0.25pts  | 4, 6, 8, 10, 16, 20 ticks        | NQ more volatile — use larger range than ES for equivalent signal quality.                  |

|                      |                  |                       |                                                                           |
|----------------------|------------------|-----------------------|---------------------------------------------------------------------------|
| CL (Crude Oil)       | \$10.00 / 0.01   | 4, 6, 8, 10, 12 ticks | High volatility. Valtos teaches CL extensively in his courses.            |
| GC (Gold)            | \$10.00 / 0.10   | 4, 6, 8, 10 ticks     |                                                                           |
| ZB (30-yr T-Bond)    | \$31.25 / 1/32   | 2, 4, 6, 8 ticks      | Lower volatility instrument — smaller range sizes appropriate.            |
| 6E (Euro FX Futures) | \$12.50 / 0.0001 | 4, 6, 8, 10, 12 ticks | Dale's primary day-trading instrument. 6-range mentioned in his examples. |
| YM (Dow futures)     | \$5.00 / 1pt     | 4, 6, 8, 10, 16 ticks |                                                                           |

***i** Range bars are Valtos's first choice because: (1) bars only form when price MOVES — eliminating time-wasting quiet periods, (2) each bar has a guaranteed price range making imbalance comparison consistent across bars, (3) order flow signals are cleaner with less noise from low-activity periods.*

## 7.4 VOLUME-Based Bar Sizes

For VOLUME bars, the user specifies the number of contracts that must trade before a new bar closes. Each bar represents approximately equal market participation regardless of time elapsed.

| Volume Size      | Config Value | When to Use                                               | Notes                                                                                           |
|------------------|--------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 100 contracts    | 100          | Thin markets; educational / development testing           | Very small — many bars per minute in active markets. Not recommended for live trading on ES.    |
| 250 contracts    | 250          | Thin futures markets (ZB, 6J)                             |                                                                                                 |
| 500 contracts    | 500          | Moderately active instruments                             |                                                                                                 |
| 1,000 contracts  | 1000         | ES, NQ standard scalping TF                               | 1000-volume bars on ES = roughly 1 bar per 10–30 seconds during active trading.                 |
| 2,500 contracts  | 2500         | ES active day trading                                     | Dale uses 2500-volume bars in some of his examples. Good balance.                               |
| 5,000 contracts  | 5000         | ES primary intraday — equivalent to roughly 5-minute time | More stable bar duration. Imbalances very reliable.                                             |
| 10,000 contracts | 10000        | ES — Valtos mentions 10000 volume chart in book           | Valtos explicitly references the 10,000 volume E-mini SP chart. Each bar represents significant |

|                  |         |                                                                                     |                                                                                                   |
|------------------|---------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
|                  |         |                                                                                     | institutional participation.                                                                      |
| 25,000 contracts | 25000   | ES — session-level macro view                                                       | Very few bars per session. Each bar tells a meaningful institutional story.                       |
| Custom           | any int | User-defined to match their instrument's typical session volume ÷ desired bar count | Formula: $\text{daily\_avg\_volume} \div \text{target\_bars\_per\_session} = \text{volume\_size}$ |

## 7.5 Multi-Chart Configuration (Recommended Workspace)

Valtos and Dale both recommend using MULTIPLE chart types simultaneously. The engine must support opening multiple bar streams per instrument. Dale's documented workspace:

| Chart Position                    | Bar Type       | Bar Size          | Purpose in Workspace                                                                                |
|-----------------------------------|----------------|-------------------|-----------------------------------------------------------------------------------------------------|
| Top Left — Context chart          | TIME           | 30 min            | Bigger picture. Bid+Ask total volume cells. Session structure, Volume Profile, Weekly VWAP context. |
| Top Right — Primary trading chart | RANGE or TIME  | 4-range or 5 min  | Main execution chart. Full footprint with bid/ask split, imbalances, COT, delta.                    |
| Bottom Left — Confirmation chart  | VOLUME or TIME | 2500 vol or 1 min | CVD divergence watch. Delta trend. Short-term momentum confirmation.                                |
| Bottom Right — Tick / DOM chart   | TIME           | 1 min or tick     | Entry precision. Live delta building. Large print detection (Trades Filter).                        |

***i** All four charts in the workspace are linked by instrument but NOT by bar type or size. Changing the instrument on one chart updates all four. The engine streams four independent bar type/size combinations simultaneously per instrument.*

## 7.6 Bar Configuration API Parameters

The `bar_type` and `bar_size` parameters apply across ALL API endpoints and WebSocket subscriptions:

| Parameter             | Type        | Valid Values        | Default | Notes                                     |
|-----------------------|-------------|---------------------|---------|-------------------------------------------|
| <code>bar_type</code> | enum string | TIME, RANGE, VOLUME | TIME    | Case-insensitive. Also accepts T, R, V as |

|                        |               |                                                                                                                    |          |                                                                                                                     |
|------------------------|---------------|--------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------|
|                        |               |                                                                                                                    |          | shorthand.                                                                                                          |
| bar_size               | string or int | See §7.2–7.4 tables. Examples: 5s, 30s, 1m, 5m, 15m, 30m, 1h, 4h, 1D, 1W, 4 (range ticks), 5000 (volume contracts) | 5m       | For TIME: use duration string. For RANGE/VOLUME: use integer.                                                       |
| session_filter         | enum string   | ALL, RTH_ONLY, ETH_ONLY, GLOBEX                                                                                    | RTH_ONLY | RTH = Regular Trading Hours. ETH = Extended. GLOBEX = full 23h session for CME futures.                             |
| timezone               | string        | Any IANA timezone. e.g. America/New_York, UTC, Asia/Tokyo                                                          | UTC      | Affects how session_open/close shorthands resolve. Does NOT change ts field format (always UTC ns).                 |
| include_incomplete_bar | bool          | true / false                                                                                                       | false    | If true, the current open (building) bar is included in historical queries. Partial data — use with caution.        |
| align_to_session       | bool          | true / false                                                                                                       | true     | If true, bar boundaries align to session open (9:30:00 ET for ES). If false, bars start from first tick of history. |

## 7.7 Bar Size Validation Rules

The engine enforces the following validation rules on bar\_size to prevent misconfiguration:

| Rule | Constraint | Error if Violated |
|------|------------|-------------------|
|------|------------|-------------------|

|                                        |                                                                         |                                                                                                                       |
|----------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| TIME minimum                           | 5 seconds (5s)                                                          | HTTP 400: bar_size_too_small. Sub-5s bars generate excessive data with minimal signal value.                          |
| TIME maximum                           | Monthly (1MO)                                                           | HTTP 400: bar_size_too_large for TIME type.                                                                           |
| TIME — must be clean divisor of 1 hour | Valid: 5m, 6m, 10m, 12m, 15m, 20m, 30m, 60m. Invalid: 7m, 11m, 13m, 17m | HTTP 400: bar_size_not_clean_divisor. Irregular sizes create misaligned bar boundaries that confuse session analysis. |
| RANGE minimum                          | 1 tick                                                                  | HTTP 400: range_too_small                                                                                             |
| RANGE maximum                          | 500 ticks                                                               | HTTP 400: range_too_large. Bars would rarely close.                                                                   |
| VOLUME minimum                         | 50 contracts                                                            | HTTP 400: volume_too_small                                                                                            |
| VOLUME maximum                         | 1,000,000 contracts                                                     | HTTP 400: volume_too_large. Bars would rarely close.                                                                  |
| RANGE on TIME chart                    | Not combinable                                                          | HTTP 400: bar_type_size_mismatch. Range size must use bar_type=RANGE.                                                 |
| Tick chart (non-standard)              | Not supported in v1.0                                                   | HTTP 400: bar_type_not_supported. Tick charts (1 bar per trade) planned for v2.0.                                     |

## 7.8 Effect of Bar Size on Order Flow Signal Quality

Bar size has a direct impact on the quality and reliability of each order flow signal. This table guides developers in selecting appropriate bar sizes for each signal type:

| Signal / Indicator  | Too Small (Noisy) | Optimal Range                         | Too Large (Sparse) | Recommended Default              |
|---------------------|-------------------|---------------------------------------|--------------------|----------------------------------|
| Footprint Imbalance | < 5s or < 2-range | 30s–5m TIME, 4–8 RANGE, 1000–5000 VOL | > 30m TIME         | 5m TIME or 4-RANGE               |
| Stacked Imbalance   | < 1m TIME         | 5m, 15m, 30m TIME                     | > 1h TIME          | 15m TIME (Valtos recommendation) |
| Delta / CVD         | < 5s              | 1m–5m TIME for intraday               | > 30m TIME         | 1m for CVD watch, 5m for signal  |

|                                     |                                |                                          |                                              |                                             |
|-------------------------------------|--------------------------------|------------------------------------------|----------------------------------------------|---------------------------------------------|
| CVD Divergence                      | < 1m TIME                      | 1m–5m TIME                               | > 15m TIME                                   | 1m TIME                                     |
| Orderflows Pulse                    | < 30s                          | 30s–5m TIME, 4–8 RANGE                   | > 15m TIME                                   | 5m TIME (Dale default)                      |
| Orderflows Turns                    | < 15s                          | 30s–3m TIME, 4-RANGE                     | > 10m TIME                                   | 30s TIME or 4-RANGE (Valtos recommendation) |
| Orderflows Ratio (Big Buyer/Seller) | < 30s                          | 1m–15m TIME                              | > 1h TIME                                    | 5m TIME                                     |
| Single Prints (Last Buyer/Seller)   | < 30s                          | 1m–30m TIME                              | > 2h TIME                                    | 5m–15m TIME                                 |
| Volume Profile POC/VAH/VAL          | N/A — VP is always accurate    | 5m–30m TIME for clear profile            | N/A                                          | 5m or 30m TIME                              |
| Volume Profile Shape (D/P/b/Thin)   | < 5m TIME (profile not mature) | 30m–1h TIME (profile mature mid-session) | N/A — shape only meaningful for full session | Check shape after 2 hours of RTH trading    |
| VWAP Reaction Signal                | < 30s (too many false touches) | 1m–5m TIME                               | > 30m TIME                                   | 5m TIME (Dale explicit)                     |
| COT (Commitment of Traders)         | < 1m TIME                      | 5m–15m TIME                              | > 1h TIME                                    | 5m TIME                                     |
| Absorption Detection                | < 1m TIME (not enough volume)  | 5m–30m TIME                              | > 2h TIME                                    | 5m–15m TIME                                 |
| Market Sweep Detector               | < 15s                          | 30s–5m TIME, 4–6 RANGE                   | > 30m TIME                                   | 1m TIME or 4-RANGE                          |
| Trapped Buyers/Sellers              | < 1m TIME                      | 5m–30m TIME                              | > 1h TIME                                    | 5m–15m TIME                                 |

 Signals generated on bars smaller than their 'Too Small' threshold will be flagged in the API response with `signal_quality=LOW`. Developers should filter these out or treat them as advisory only.

## 7.9 Bar Size in WebSocket Subscriptions

When subscribing to live streams, `bar_type` and `bar_size` are encoded in the topic string:

```
SUBSCRIBE footprint:ES:TIME:5m // 5-minute TIME bars for ES
```



```

SUBSCRIBE footprint:ES:RANGE:4           // 4-tick RANGE bars for ES
SUBSCRIBE footprint:ES:VOLUME:5000       // 5000-contract VOLUME bars for ES
SUBSCRIBE delta:live:ES:TIME:1m          // live delta on 1-minute bars
SUBSCRIBE pulse:ES:TIME:5m               // Pulse signals on 5-minute bars
SUBSCRIBE turns:ES:RANGE:4               // Turns signals on 4-range bars
SUBSCRIBE imbalances:ES:TIME:15m         // Imbalances on 15-minute bars
SUBSCRIBE profile:live:ES:TIME:5m        // Live session Volume Profile, 5-min
resolution

```

**i** A developer can subscribe to **MULTIPLE** bar sizes simultaneously for the same instrument. Example: subscribe to `pulse:ES:TIME:5m` AND `pulse:ES:TIME:30m` to implement multi-timeframe confluence detection in their own code.

## 7.10 Bar Size in REST Historical Queries

All historical endpoints accept `bar_type` and `bar_size` as query parameters:

```

GET /v1/footprint/ES?bar_type=TIME&bar_size=5m&from_ts=session_open&to_ts=now
GET /v1/footprint/ES?bar_type=RANGE&bar_size=4&from_ts=2026-05-30T09:30:00Z
GET /v1/signals/pulse/ES?bar_type=TIME&bar_size=5m&from_ts=now-5d&to_ts=now
GET /v1/signals/stacked_imbalance/ES?bar_type=TIME&bar_size=15m&from_ts=now-30d
GET /v1/profile/ES/session?bar_type=TIME&bar_size=5m&session_date=2026-05-30

```

**i** Historical data is stored at the **TICK** level and re-aggregated into any requested `bar_type`/`bar_size` on the fly. This means a developer can query any bar size from history without that exact size having been pre-computed. Re-aggregation adds ~50–200ms to query time for large date ranges.

## 8. Timestamp Specification — Signal vs Snapshot

### 8.1 The Fundamental Difference

| Timestamp Type     | One-Line Definition                                                  | Triggered By                                                         | Trading Significance                          |
|--------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------|
| Signal Timestamp   | The exact nanosecond a specific market condition BECAME TRUE         | A market event — a condition crossing a threshold for the first time | HIGH — marks the birth of an actionable setup |
| Snapshot Timestamp | The exact nanosecond a periodic reading of an indicator was RECORDED | A timer or bar close — regardless of whether the value changed       | LOW — reflects current state, not a new event |

### 8.2 Side-by-Side Architecture

| Property               | Signal Timestamp                                             | Snapshot Timestamp                                     |
|------------------------|--------------------------------------------------------------|--------------------------------------------------------|
| Subsystem              | Signal Detection Engine — event-driven                       | Metric Streaming Engine — timer-driven                 |
| Trigger                | Condition threshold crossed for first time                   | Timer fires (100ms, 5s) OR bar closes                  |
| Frequency              | Sparse — minutes or hours between signals on same instrument | Dense — tens to hundreds per minute                    |
| Deduplication          | YES — same signal not re-emitted until condition resets      | NO — snapshots repeat even if value unchanged          |
| Value change required? | YES — fires only when something NEW happens                  | NO — emits even if value is identical to last snapshot |
| Use in API query       | Filter: 'show me signals BETWEEN these two times'            | Filter: 'show me the indicator VALUE AT this time'     |
| Persisted in DB?       | YES — every signal permanently stored with full context      | NO — latest in Redis cache; history via bar OHLCV      |
| WebSocket behavior     | Fires once per event. Quiet between events.                  | Fires on every timer tick. Continuous stream.          |

### 8.3 The Four Signal Timestamp Fields

Every signal record contains four timestamp fields — each with a distinct meaning:

| Field | What It Records | When to Use |
|-------|-----------------|-------------|
|-------|-----------------|-------------|

|              |                                                                                                                                 |                                                                                                  |
|--------------|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| detection_ts | Nanosecond the engine DETERMINED the condition was met. Always after bar close for bar-close signals. Primary signal timestamp. | Strategy logic. 'When did I know about this signal?' Use in backtesting as the action timestamp. |
| bar_open_ts  | Nanosecond the BAR that triggered the signal OPENED. Always earlier than detection_ts.                                          | Bar attribution. 'Which bar caused this?' Use to align signal with chart data.                   |
| bar_close_ts | Nanosecond the triggering bar CLOSED. For bar-close signals ≈ detection_ts.                                                     | Execution timing. 'Enter at open of the bar AFTER this timestamp.'                               |
| exchange_ts  | Exchange-assigned timestamp of the LAST TICK that completed the condition. Most precise market-event time.                      | Regulatory / audit use. Matches exchange records exactly.                                        |

**These four always follow this strict ordering:**

```
exchange_ts <= bar_open_ts < bar_close_ts <= detection_ts
```

## 8.4 All Valid Timestamp Formats

| Format                          | Example                        | Notes                                                                              |
|---------------------------------|--------------------------------|------------------------------------------------------------------------------------|
| Unix nanoseconds (int64)        | 1748599800000000000            | Preferred. 19 digits. Unambiguous. Fastest to parse. Use in all production code.   |
| Unix milliseconds (int64)       | 1748599800000                  | 13 digits. Auto-detected by digit count. Converted internally to ns (× 1,000,000). |
| Unix seconds (int64 or float64) | 1748599800 or 1748599800.291   | 10 digits integer or float with fractional seconds.                                |
| ISO 8601 UTC                    | 2026-05-30T09:35:00Z           | Z suffix REQUIRED. Without it: HTTP 400 missing_timezone.                          |
| ISO 8601 UTC with nanoseconds   | 2026-05-30T09:35:00.000003847Z | Up to 9 decimal places accepted.                                                   |
| ISO 8601 with timezone offset   | 2026-05-30T04:35:00-05:00      | Offset accepted. Internally converted to UTC.                                      |
| Date-only string                | 2026-05-30                     | Expands to 2026-05-30T00:00:00.000000000Z (midnight UTC).                          |
| Relative: now                   | now                            | Current server UTC time at query execution.                                        |
| Relative: now-Xs                | now-30s, now-45s               | X seconds ago. Min 1s.                                                             |
| Relative: now-Xm                | now-15m, now-30m               | X minutes ago.                                                                     |

|                    |                    |                                                                                |
|--------------------|--------------------|--------------------------------------------------------------------------------|
| Relative: now-Xh   | now-4h, now-6h     | X hours ago.                                                                   |
| Relative: now-Xd   | now-1d, now-5d     | X calendar days ago.                                                           |
| Relative: now-Xw   | now-2w, now-4w     | X weeks ago.                                                                   |
| Relative: now-XM   | now-3M, now-6M     | X months ago (capital M = months).                                             |
| Relative: now-Xy   | now-1y, now-2y     | X years ago.                                                                   |
| session_open       | session_open       | Today's RTH session open for the queried instrument (e.g. 09:30:00 ET for ES). |
| session_close      | session_close      | Today's RTH session close (e.g. 16:00:00 ET for ES).                           |
| prev_session_open  | prev_session_open  | Previous trading day's session open.                                           |
| prev_session_close | prev_session_close | Previous trading day's session close.                                          |
| week_open          | week_open          | Monday's session open for the current trading week.                            |
| month_open         | month_open         | First trading day of the current calendar month, session open.                 |
| year_open          | year_open          | First trading day of current calendar year, session open.                      |
| epoch              | epoch              | 2010-01-01T00:00:00Z — fetch the maximum available history.                    |

## 8.5 Valid vs Invalid — Quick Examples

✓ 1748599800000000000 — Unix nanoseconds, 19 digits, within range  
 ✓ 2026-05-30T09:35:00Z — ISO 8601 UTC with Z suffix  
 ✓ 2026-05-30T04:35:00-05:00 — ISO with offset (same UTC moment)  
 ✓ now-4h — valid relative shorthand  
 ✓ session\_open — session-relative shorthand  
 ✓ 1748599800.291 — Unix seconds with fractional part  
 ✓ epoch — fetch all history

✗ 2026-05-30T09:35:00 — INVALID: missing timezone suffix → HTTP 400 missing\_timezone  
 ✗ 05/30/2026 09:35:00 — INVALID: US date format not supported → HTTP 400 invalid\_timestamp\_format  
 ✗ now+1d — INVALID: future timestamps not allowed → HTTP 400 timestamp\_in\_future  
 ✗ 0 — INVALID: Unix epoch zero predates all data → HTTP 400 timestamp\_too\_early  
 ✗ 1748599800000 — AMBIGUOUS: 13 digits treated as milliseconds, not nanoseconds — be explicit

## 8.6 Signal Timestamp Nuances by Signal Type

| Signal Type          | detection_ts Behavior                                             | Important Developer Note                                                                                   |
|----------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Stacked Imbalance    | Set when Nth consecutive imbalance is detected at bar close       | zone_id persists. Extensions don't generate new detection_ts. Use zone_created_ts vs zone_last_updated_ts. |
| Orderflows Pulse     | Set at bar close when composite score first crosses threshold     | Can re-fire: if score drops then rises again, new detection_ts on new bar.                                 |
| Orderflows Turns     | Set at TICK level when turning pattern completes — not bar close  | exchange_ts ≈ detection_ts within microseconds. Most precise signal.                                       |
| Orderflows Ratio     | Set at bar close when bid/ask ratio at extreme crosses threshold  | One signal per bar maximum. Consecutive qualifying bars each get own detection_ts.                         |
| Single Print         | Set at bar close when zero/near-zero volume at extreme detected   | magnet_active=true persists until resolved. magnet_deactivation_ts records resolution.                     |
| Delta Surge          | Set when surge_magnitude first crosses threshold (every 100ms)    | detection_ts = first crossing. surge_peak_ts = when surge peaked.                                          |
| CVD Divergence       | Set when divergence confirmed for N consecutive bars              | detection_ts = Nth bar confirmation. divergence_start_ts = when pattern began.                             |
| VWAP Reaction        | Set at TICK level when price touches VWAP and order flow confirms | Mid-bar signal — entry may be possible on same bar. Use exchange_ts for precise entry timing.              |
| Volume Profile Shape | Set at each bar close as shape evolves during session build       | Check shape_confidence field. Low confidence early in session — shape may change.                          |

## 8.7 Timestamp Query Quick Reference

| I want to...                     | Parameter            | Example                                                  |
|----------------------------------|----------------------|----------------------------------------------------------|
| Get signals today                | from_ts=session_open | GET /v1/signals/pulse/ES?<br>from_ts=session_open        |
| Get signals in specific hour     | from_ts + to_ts      | ?from_ts=2026-05-30T14:00:00Z&to_ts=2026-05-30T15:00:00Z |
| Identify which bar caused signal | bar_close_ts field   | Read bar_close_ts from signal response                   |
| Find entry timing after signal   | bar_close_ts + 1 bar | Enter at open of bar AFTER bar_close_ts                  |

|                                     |                                          |                                                           |
|-------------------------------------|------------------------------------------|-----------------------------------------------------------|
| Get current VWAP                    | latest=true                              | GET /v1/vwap/ES?latest=true                               |
| Get VWAP at specific moment         | at_ts                                    | GET /v1/vwap/ES?at_ts=2026-05-30T14:00:00Z                |
| Run backtest last 6 months          | from_ts=now-6M                           | POST /v1/backtest { from_date: 'now-6M', to_date: 'now' } |
| Get only RTH session signals        | from_ts=session_open&to_ts=session_close | Excludes pre/post market                                  |
| Get unresolved Single Print magnets | resolved=false                           | GET /v1/signals/single_prints/ES?resolved=false           |
| Get last 5 Pulse signals            | order+limit                              | ?<br>order_by=detection_ts&direction=desc&limit=5         |
| Replay missed signals on WS connect | since=now-1h                             | SUBSCRIBE pulse:ES?since=now-1h                           |

---

### **END OF SPECIFICATION — v3.0**

*v3.0 additions: Section 7 — Bar & Candlestick Time Configuration | Section 8 — Timestamp Specification*